

The IMAT Syllabus

Every topic across Biology, Chemistry, Physics & Maths, and Logic & Reading, organized the way we actually teach it, not just the way the decree lists it.

Source: Decreto Ministeriale (MUR), Allegato 1

Official syllabus programme

imat.club

Club AI, Pro-only

This is our own organization of the official syllabus, offered as guidance only and not official or legal advice. The Ministry's decree is the binding source, always verify current details yourself.

Format & scoring

Everything on the following pages is scoped to what these 60 questions actually draw from, current CINECA/University format.

Total questions	60
Time allowed	100 minutes
Options per question	5 (A–E), one correct
Correct answer	+1.5
Wrong answer	–0.4
Unanswered	0
Maximum score	90

Section breakdown

Reading Skills & Knowledge Acquired in Studies	4 questions
Logical Reasoning & Problems	5 questions
Biology	23 questions
Chemistry	15 questions
Physics & Mathematics	13 questions

Section weightings and scoring are stable year to year, per the official decree. The topic lists on the following pages are our own grouping of the same official programme, organized for how it's actually studied, not necessarily the decree's own section order.

Biology

23 of 60 questions, nearly double the size of any other section.

Genetics & Heredity

Mendelian inheritance & ratios, Pedigree analysis, Allele frequency & Hardy-Weinberg, Mutations & their effects, Chromosomal inheritance

Cell Biology

Organelle structure & function, Membrane transport, Cell communication & signaling, Cell cycle & division, Cytoskeleton

Bioenergetics

Glycolysis, Krebs cycle, Oxidative phosphorylation, Photosynthesis basics, ATP & energy currency

Molecular Biology

DNA replication, Transcription, Translation, Gene regulation, Mutations at the molecular level

Human Anatomy & Physiology

Cardiovascular system, Respiratory system, Nervous system, Digestive system, Endocrine system, Musculoskeletal system

Evolution & Ecology

Natural selection mechanics, Speciation, Population genetics, Ecological relationships & food webs, Biodiversity & conservation

Chemistry

15 of 60 questions. Formula-application over pure theory.

Organic Chemistry

Functional groups, Nomenclature basics, Hydrocarbons, Isomerism, Common reaction types

Acids, Bases & pH

pH & pOH calculations, Strong vs weak acids/bases, Titration basics, Buffers, K_a & K_b

Electrochemistry

Oxidation states, Redox reactions, Galvanic cells, Electrolysis basics

Stoichiometry & Reactions

Mole calculations, Limiting reagent, Percent yield, Balancing equations, Concentration & dilution

Atomic Structure & Bonding

Electron configuration, Periodic trends, Ionic vs covalent bonding, Molecular geometry, Intermolecular forces

Thermochemistry & Kinetics

Exothermic vs endothermic reactions, Enthalpy & Hess's law, Reaction rates, Activation energy, Equilibrium basics

Physics & Maths

13 of 60 questions, split across physics and maths.

Dynamics & Energy

Newton's laws, Work & energy, Momentum & impulse, Circular motion

Electrostatics & Circuits

Coulomb's law, Electric field & potential, Ohm's law, Series & parallel circuits

Fluid Mechanics

Density & pressure, Pascal's principle, Archimedes' principle, Bernoulli's principle

Probability & Geometry

Basic probability, Ratios & proportions, Area & volume, Reading graphs & data

Kinematics

Uniform motion, Uniformly accelerated motion, Projectile motion, Free fall

Thermodynamics

Temperature & heat, Gas laws, Laws of thermodynamics, Specific heat & calorimetry

Algebra & Functions

Solving equations, Rearranging formulas, Exponents & scientific notation, Basic functions & graphs

Logic & Reading

9 of 60 questions across logical reasoning and reading comprehension.

Logical Problems

Propositional logic, Deductive reasoning, Seating/arrangement puzzles, Syllogisms

Argument Analysis

Identifying assumptions, Strengthening/weakening arguments, Spotting logical fallacies

General Culture & History

World history milestones, Geography basics, Art & literature, Current affairs

Numerical & Data Reasoning

Percentages & ratios, Data table interpretation, Basic statistics, Estimation under time pressure

Reading Comprehension

Main idea & inference, Vocabulary in context, Author's tone & purpose, Literary vs informational passages

Institutions & Citizenship

Government structures, Italian & EU institutions, Rights & civic duties, Separation of powers



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